

Robustness, Cascades and Bus Recovery in the Transperth Network

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Abstract

We model robustness, overload cascades and bus recovery on an audited 86-station Transperth railway. Targeted removal fragments the network sooner than random station loss. Local load redistribution produces abrupt secondary failure, whereas shortest-path recomputation on the rail tree contains overload by dropping unreachable journeys. Bus links can restore service while increasing rail failures; with a two-link policy after an Airport Line closure, served terminal pairs fall from 93.11% to 66.68%. Sampling and grid checks qualify the tolerance thresholds, and the finite avalanche samples do not support a power-law claim. These findings concern specified routing and capacity proxies, rather than measured passenger behaviour or operational deployment advice.

1 Introduction

1.1 Background and research aims

A station closure can sever a railway corridor before any further facility fails. Substitute buses may reconnect journeys, but the paths they restore can also concentrate traffic at surviving rail facilities. We investigate these competing effects in the Transperth network, distinguishing infrastructure connectivity, secondary overload failures and the fraction of terminal pairs that remain served. This distinction matters because fewer operational rail facilities need not imply less travel service when another transport layer is available.

Albert et al. (2000) established the distinction between random errors and targeted removal in heterogeneous networks. Their scale-free-network results motivate a comparison of failure strategies here; they do not establish that Transperth has the same degree distribution or resilience. Our audited railway has 86 mapped stations and 85 adjacent segments and is a tree. It therefore lacks alternative rail paths around a removed station. Its inputs represent Monday 5 October 2026, 07:00–09:00 in Australia/Perth, with 271 selected rail trips and 4,324 selected bus trips. These are scheduled services, not measured passenger journeys. Showgrounds remains a mapped station although it has no weekday stop event in this window.

Motter and Lai (2002) showed how a local attack can cause further overload failures when shortest-path loads are recomputed against capacities fixed from the intact network. We use that capacity construction and compare it with a local redistribution scenario in which a failed facility forwards its carried load to surviving neighbours. These rules represent different assumptions about disrupted travel. Their comparison tests whether an apparent tolerance transition depends on the update mechanism. In the bus layer, restored paths can increase surviving rail loads even while improving terminal reachability. Apparent avalanche scaling requires a separate statistical test: following Clauset et al. (2009), a visual slope alone is not evidence for a power law.

Table 1: Scenarios, parameter ranges and outcome measures.

Scenario	Parameters	Outcome measures
Baseline	values	measures
Variation A	values	measures

Our research questions are: **RQ1**, how do random and targeted removal fragment the railway, and which station characteristics identify bottlenecks? **RQ2**, how do capacity tolerance and load-update rules affect the size and duration of secondary failures? **RQ3**, when do bus backup links improve served travel, and can their rerouting aggravate a rail cascade? **RQ4**, how sensitive are these conclusions to sampling, tolerance-grid resolution and the assumed load/capacity model, and is avalanche scaling statistically supported?

1.2 Contribution

Our contribution is a reproducible transport case study that joins an audited GTFS-derived topology to removal, overload and recovery experiments. We expand express-service traversals over mapped adjacent segments rather than treating skipped stops as shortcuts. We separate persistent passenger terminals from closable rail facilities, retain evidence for substitute links, and compare infrastructure damage with service outcomes. Paired scenarios and bootstrap intervals expose uncertainty without treating scheduled departures as observed demand. The capacity law and robustness questions come from existing work; the audited Transperth construction, explicit comparison of load-update assumptions, and evidence-linked recovery analysis are the team’s extension. They are network experiments, not a replication of a taught agent-based model.

2 Model specification

Define the entities, state variables and interactions of the model. The update rule for a generic entity is

$$x_i(t+1) = f(x_i(t), \{x_j(t)\}_{j \in \mathcal{N}_i}; \theta), \quad (1)$$

where $\mathcal{N}_i$ is the neighbourhood of entity i and θ collects the model parameters. Explain each term and justify the form of f , referring to the implementation in `src/` of the main repository.

2.1 Assumptions and initialisation

State the modelling assumptions, boundary cases and initialisation rules, and justify the parameter ranges that are used in the experiments.

3 Experimental design

Describe the experiments used to answer the research question: scenarios, parameter sweeps, random seeds, number of repetitions and outcome measures. Summarise them in a table such as Table 1.

Explain the analysis pipeline that turns raw simulation output into the reported statistics, and point to the notebooks and scripts that implement it.

4 Results

4.1 RQ1: targeted removal exposes corridor bottlenecks

Figure 1a shows earlier fragmentation under targeted station removal. The interpolated removal fraction at which the mean largest component reaches half of the intact 86 stations is 6.32% for random removal, 1.68% for a fixed degree ranking and 1.24% for a fixed betweenness ranking. The random curve uses 100 trials per sampled fraction. These crossing estimates are distinct from the corresponding second-largest-component susceptibility peaks at 6%, 2% and 1%. They also depend on integer station-count rounding and grid resolution. On this small tree, neither diagnostic establishes a thermodynamic phase transition. Betweenness identifies corridor-separating positions that degree alone cannot distinguish: many internal stations have degree two but sit between different amounts of the network.

4.2 RQ2: tolerance depends on the load-update assumption

For the maximum-load trigger, Oats Street, static capacity-weighted redistribution at $\alpha = 0.2$ closes all 86 stations, including the trigger, in 31 secondary rounds. On the 0.025 grid, the first sampled tolerance containing the closure is $\alpha = 0.525$; equal redistribution requires $\alpha = 0.625$ (Figure 1b). These are sampled thresholds for the specified trigger and rule, not universal station-capacity requirements. Dynamic unweighted betweenness produces no secondary failures on the frozen tree: deletion cannot create a new rail route, so surviving path loads cannot increase. The resulting disconnected graph can still lose service. The 2,000 random-trigger observations at each fixed avalanche condition include zero secondary failures; rounds within one cascade are not separate trials. The observed size distribution is concentrated at a few finite outcomes (Figure 3b), rather than a broad scaling regime.

4.3 RQ3: service recovery and rail damage can move in opposite directions

After a Bayswater closure at $\alpha = 0.2$, the rail-only scenario serves 67.00% of the original 3,655 terminal pairs. Full candidate bus standby serves 100.00%, but closes two rail facilities instead of one (Figure 2a). At the first load check, Fremantle’s load-to-capacity ratio rises from 0.68 without buses to 1.31 with the full pool (Figure 2b). It exceeds the fixed rail-only capacity and fails next. Bus paths still keep its terminal and the other terminals connected. At $\alpha = 1.0$, the full-pool Bayswater case serves all pairs with only the trigger closed. Additional capacity can therefore contain this rerouting overload without being responsible for the restored paths themselves.

The complete candidate pool does not worsen served service in the stored scan; a budgeted strategy can. After the three exclusive Airport Line facilities close at $\alpha = 0.2$, the existing-bus policy selects two links and additionally closes West Leederville. Served terminal pairs fall from 93.11% with no backup to 66.68%. Endpoint-weighted served demand likewise falls from 94.50% to 69.65% (Table 2). These weights are sums of scheduled stop counts at pair endpoints, not observed OD demand. Travel-time penalties are conditional on reachable pairs; comparing their means alone could mistake loss of long journeys for faster service.

4.4 RQ4: numerical uncertainty and model sensitivity

The tested nested-prefix criteria recommend 1000 random trials: the worst normalised mean change is 0.0030 and the largest pointwise 95% interval half width is 0.0220, below the chosen 0.02 and 0.03 tolerances (Figure 3a). These criteria apply to the stored conditions, not every transport model. Coarsening the capacity-rule alpha grid from 0.025 to 0.1 shifts first sampled containment from 0.525 to 0.600. The demand/frequency-capacity scenario gives 0.750; it changes the load proxy, capacity reference and maximum-load trigger, so it is a model comparison rather

than an isolated effect of passenger demand. Frequency weighting alone leaves unique tree paths unchanged.

The bounded decreasing power-law test follows maximum-likelihood fitting, cutoff selection and refitted simulation, rather than log-log regression (Clauset et al., 2009). At static alpha 0.1–0.2 there are only two distinct positive secondary sizes, making the tail ineligible under the three-size requirement. At 0.3 the fitted hypothesis is rejected ($p = 0.002$, 500 refitted simulations), with its exponent at the lower search boundary. Dynamic samples contain only zeros. The experiments therefore do not support a power-law avalanche claim.

5 Discussion and conclusions

The tree topology explains RQ1’s vulnerability: removing a station can cut the only rail path between corridors. Extra capacity cannot restore a deleted edge. RQ2 instead concerns whether surviving facilities acquire further overloads. Static redistribution keeps forwarding a proxy quantity locally; dynamic betweenness removes contributions from unreachable pairs. Their sharp contrast reveals the importance of how disrupted journeys are represented. Interpreting the dynamic result as uninterrupted operation would confuse disappearing load with preserved service. Likewise, the static all-station cascade is conditional on the forwarding rule, not evidence that a single real closure would destroy Perth’s railway.

RQ3 shows why recovery needs both service and facility metrics. Backup paths can reconnect terminals while increasing load at particular rail facilities; the Fremantle case supplies a traced overload mechanism. Enough substitute connectivity can maintain travel despite that extra closure, whereas a limited selection can leave the network worse after secondary failures. Policies should therefore be compared after the cascade stabilises, with a common intact OD denominator and the reachable population reported beside travel times. Our two-link budget counts selected connections, not physical buses or fleet cost.

Uncertainty remains broader than the Monte Carlo intervals in RQ4. We study one service date and window, mapped undirected links, approximate rail speeds, proxy capacities, symmetric effective bus times and immediate unlimited bus standby. The model omits measured boarding/alighting, vehicle occupancy, directional timetables, congestion, dispatch delay and finite substitute capacity. Terminal-pair reachability is not proof of a feasible scheduled journey. The finite 86-station support also limits avalanche-tail inference; the rejected bounded hypothesis cannot establish the absence of scaling in a larger or different transport system.

The most useful next tests would introduce observed OD demand and vehicle capacity, directional time-dependent routing, delayed finite bus deployment, and several service-day snapshots. Enumerating all 86 deterministic station triggers would remove trigger-sampling error, while confidence intervals across dates would address a different source of uncertainty. The present conclusions are narrower: targeted corridor removal is structurally damaging; overload containment depends on the update rule and tolerance grid; and substitute connectivity can improve service while aggravating rail overload. These are reproducible mechanisms to test against operational evidence, rather than a recommendation to deploy a particular bus policy.

Selected figures

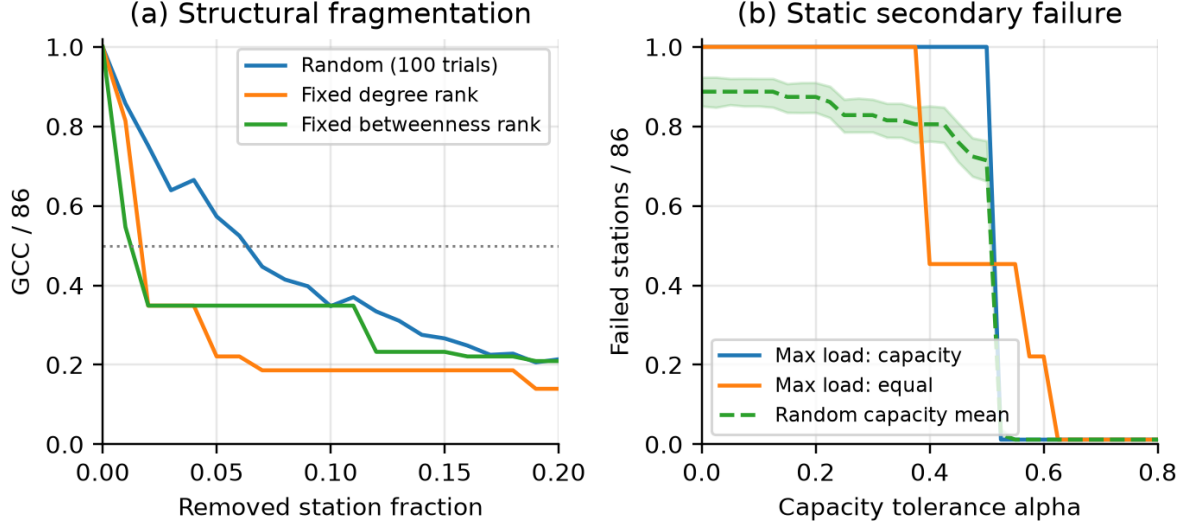


Figure 1: Frozen 86-station railway. (a) Station-removal curves, with the horizontal $\text{GCC}/86=0.5$ diagnostic. The random mean uses 100 trials per fraction; target orders are fixed from the intact graph. (b) Static unweighted-betweenness cascades. Targeted curves use the maximum-load station; the random capacity-rule mean uses 300 whole trigger trials per alpha and pointwise 95% bootstrap intervals with 2,000 resamples. Failed fractions include the trigger. Sources: P2.1 percolation tables, P2.2 runs and P2.5 cascade intervals; selected by `report/generate_figures.py`.

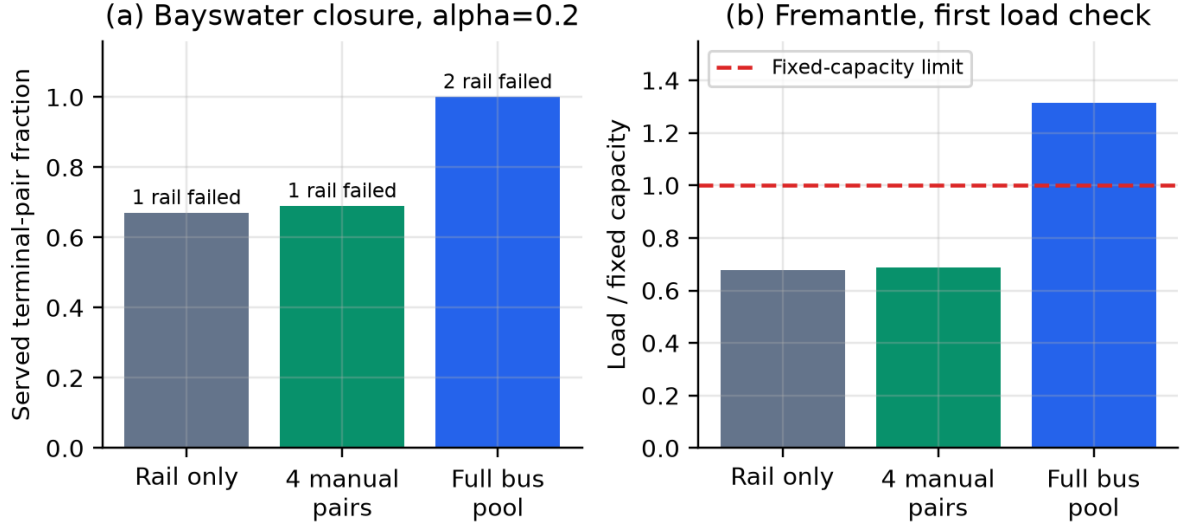


Figure 2: Bayswater closure at $\alpha = 0.2$, layered dynamic routing with capacities fixed from the intact rail-only baseline. (a) Served terminal pairs over the original 3,655 pairs; annotations count failed rail facilities, including the trigger. (b) Fremantle's load/capacity ratio immediately after the trigger, before secondary closures. The full bus pool restores all pairs while pushing Fremantle above one. This is immediate unlimited standby, not a two-link budget. Sources: P2.3 bus-cascade scan and loads tables.

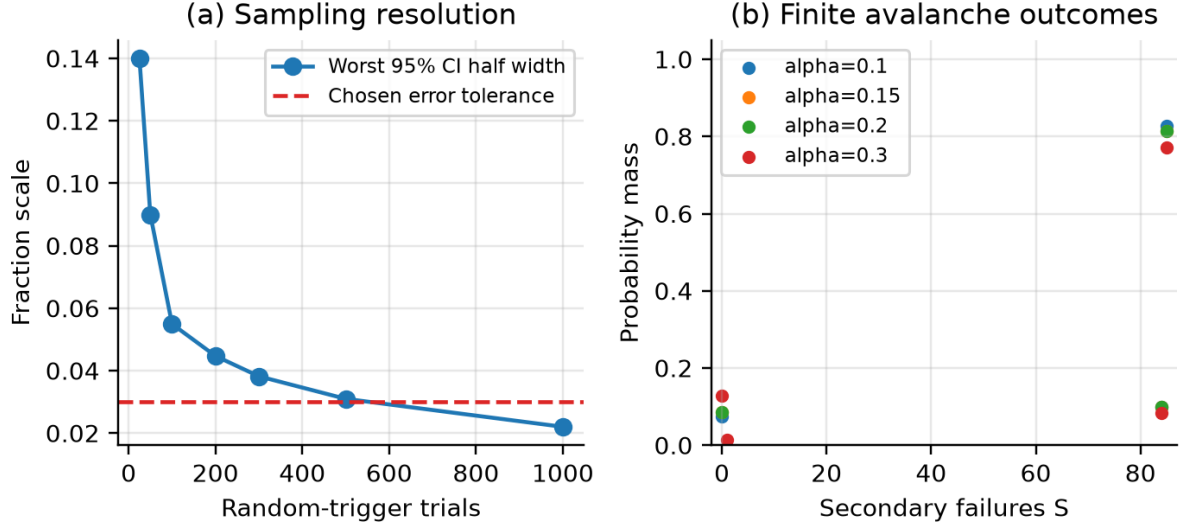


Figure 3: Uncertainty is conditional on the frozen graph and load model. (a) Worst pointwise 95% interval half width over the tested conditions with a common available prefix size. The 0.03 line is a chosen numerical tolerance; the accompanying mean-change criterion is checked separately. (b) Static capacity-rule probability masses from 2,000 random triggers at each alpha; zero secondary failures remain in the denominator and alphas are not pooled. Most mass lies at a few finite sizes, unsuitable evidence for broad scaling. Sources: P2.5 recommendations and P2.2 avalanche samples.

References

- Réka Albert, Hawoong Jeong, and Albert-László Barabási. Error and attack tolerance of complex networks. *Nature*, 406:378–382, 2000. doi: 10.1038/35019019. Correction: *Nature* 409, 542 (2001), doi:10.1038/35054111.
- Aaron Clauset, Cosma Rohilla Shalizi, and M. E. J. Newman. Power-law distributions in empirical data. *SIAM Review*, 51(4):661–703, 2009. doi: 10.1137/070710111.
- Adilson E. Motter and Ying-Cheng Lai. Cascade-based attacks on complex networks. *Physical Review E*, 66:065102, 2002. doi: 10.1103/PhysRevE.66.065102.

A Reproducibility and supplementary evidence

The repository is <https://github.com/CITS4403-Project/Project>. The live demonstration uses `notebooks/01_network.ipynb` and `notebooks/03_cascades.ipynb`, alongside the P3.2/P3.4 notebooks. These run from committed inputs and do not require a live timetable download. Full model definitions and raw-feed reconstruction are in `docs/model.md` and `docs/data.md`.

The RQ1 crossing table is `percolation_critical_fractions.csv` in `results/percolation/`; susceptibility peaks and GCC crossings occupy separate columns. RQ2 uses `results/cascade/runs.csv` and `avalanche_samples.csv`. RQ3 uses `results/recovery/bus_cascade_scan.csv`, `bus_cascade_loads.csv` and `strategy_comparison.csv`. RQ4 uses `results/uncertainty/recommendations.json`, `power_law_fits.json` and `results/demand/demand_alpha_star.csv`. Every source has a metadata sidecar. The selected-figure manifest additionally records table, sidecar, generator and output hashes.

From the repository root, `make figures` regenerates the three selected images and the numeric macros used in the abstract and results. Then run `make` inside `report/`, or use the documented Tectonic command in `report/README.md`. The P2 runners reproduce full experiments; the figure generator only reads their frozen outputs. Bootstrap intervals quantify pointwise Monte Carlo error; they do not cover different dates, measured demand or model misspecification.

Table 2: Airport Line exclusive-station closure, $\alpha = 0.2$. The policy budget is two selected links. Fractions use common intact denominators.

Policy	Secondary closures	Served OD (%)	Served weight (%)
No backup	0	93.11	94.50
Existing bus	1	66.68	69.65